SOMAIYA
VIDYAVIHAR
K J Somaiya Institute of Technology
An Autonomous Institute Permanently Affiliated to the University of Mumbai

DEPARTMENT OF INFORMATION TECHNOLOGY

Course Name and Code: Data Structures Lab **(**ITL302)
Semester: III (SYIT)
Academic Year: 2026-27 (Odd Semester)
Name: Rashi Dawda
Roll No: 23

Experiment No. 05

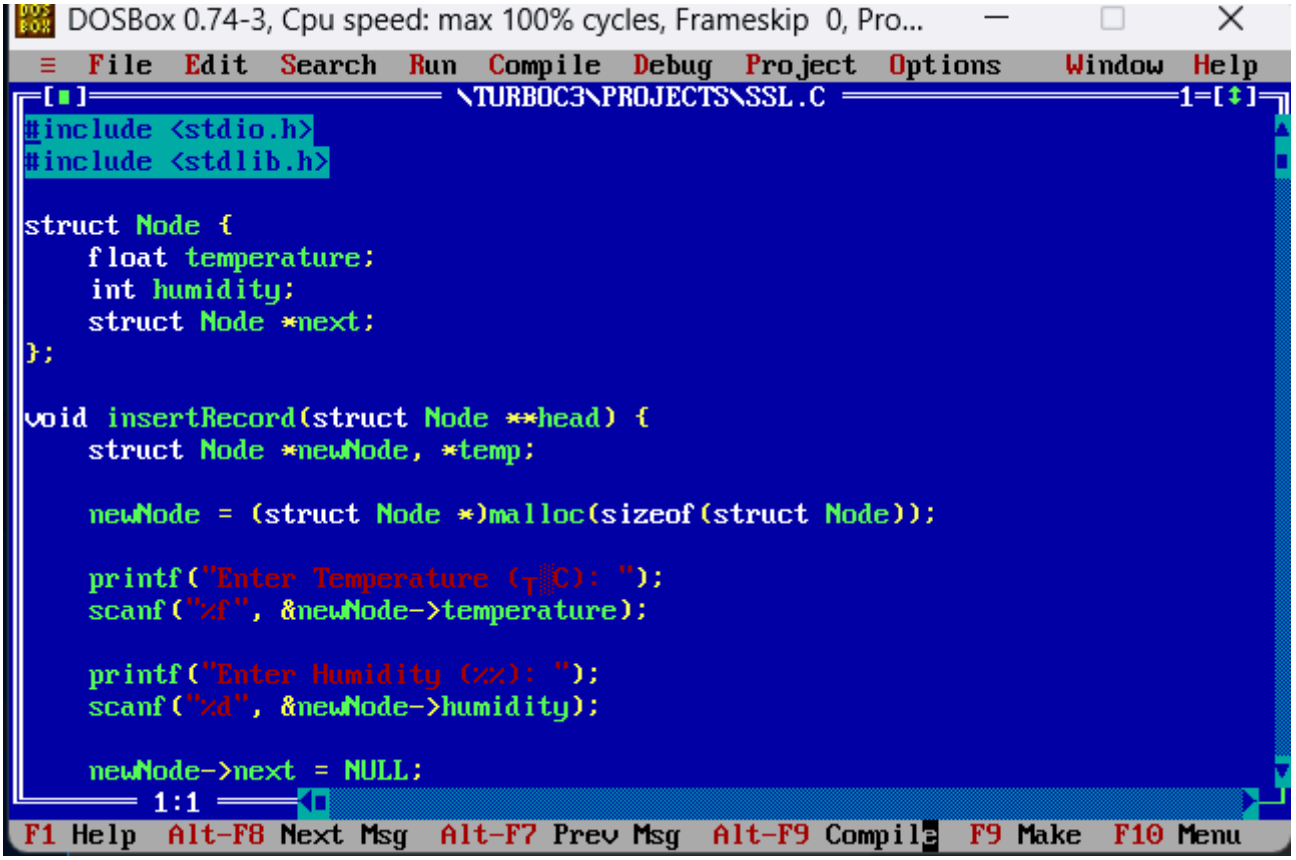
Aim: To implement a Singly Linked List with insertion, display, searching, and deletion operations for a heatwave monitoring system.

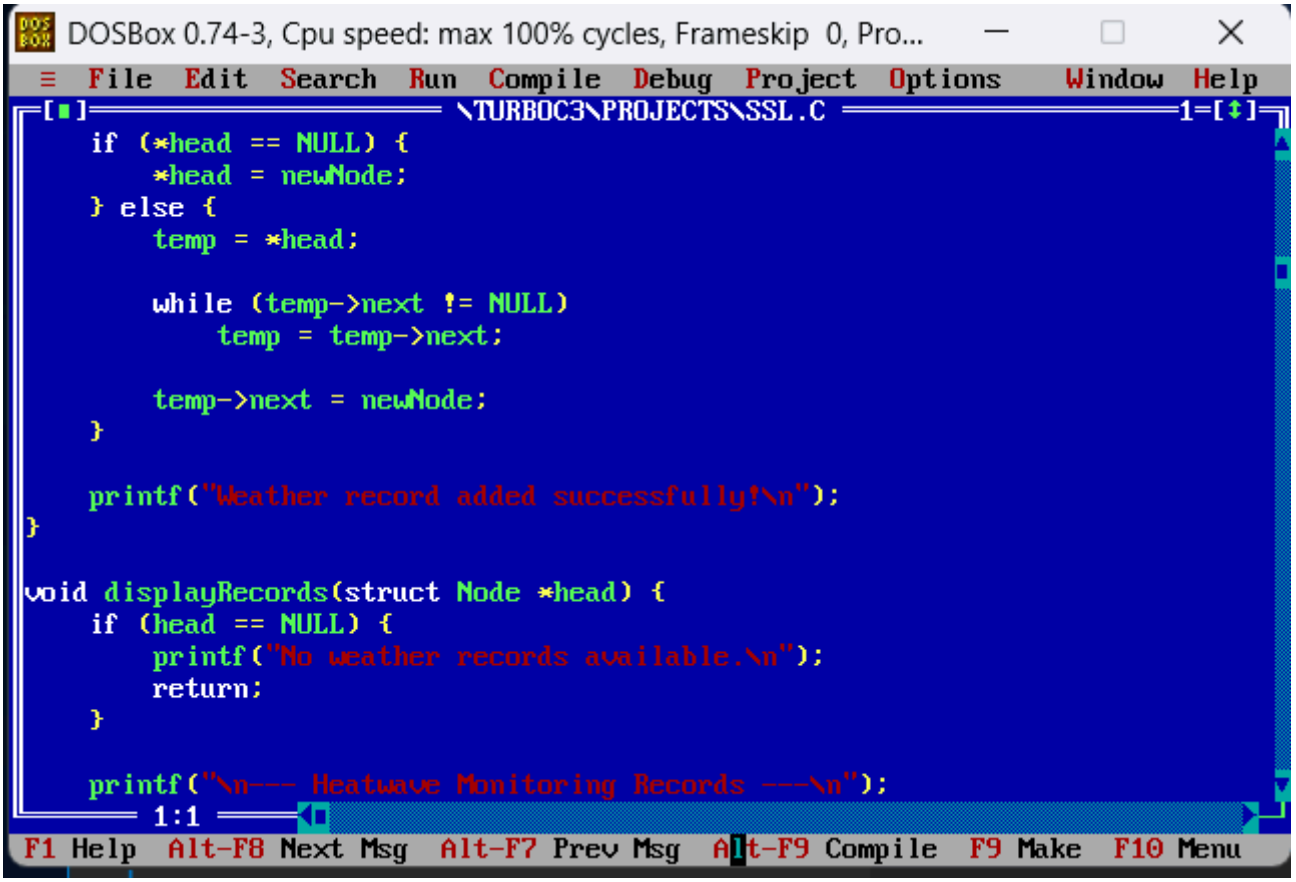
Objectives:

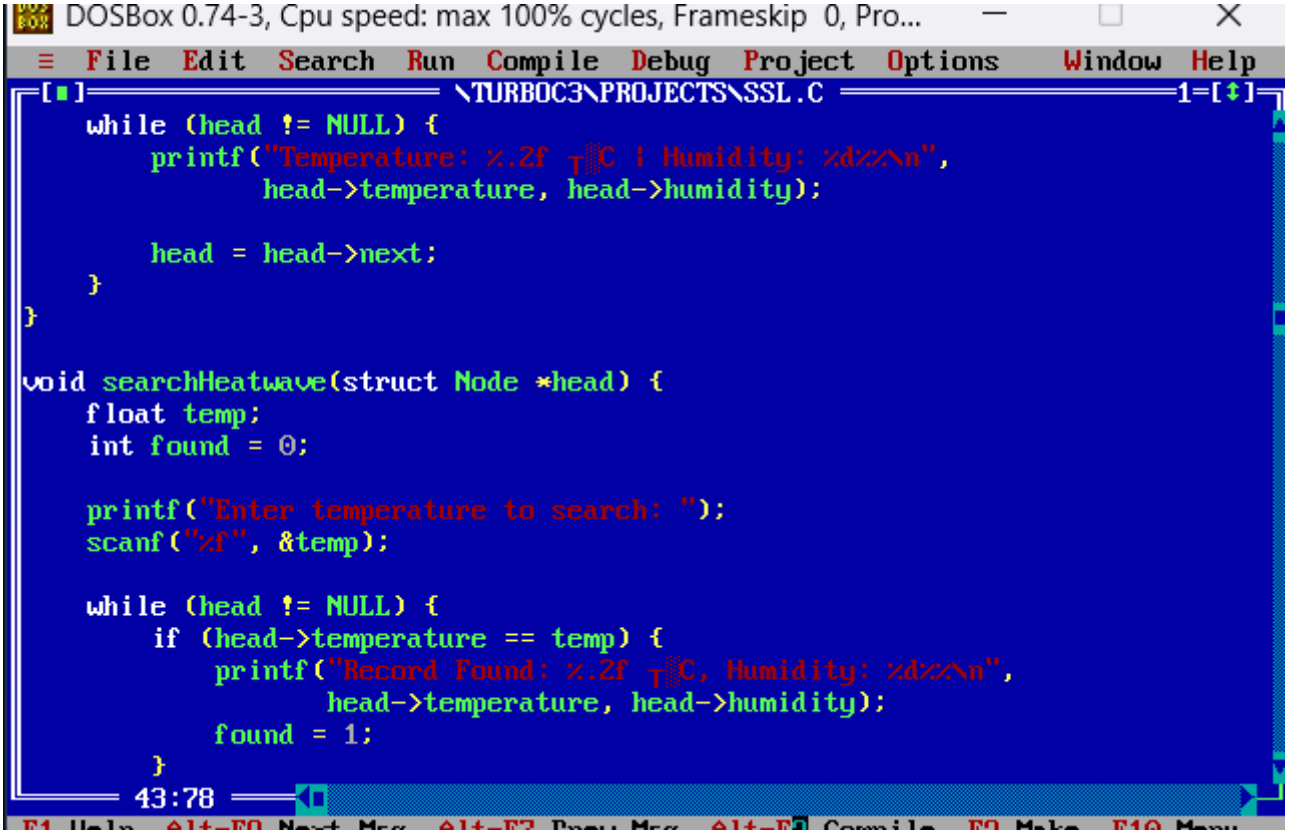
1. Understand the concept of a singly linked list.
2. Store temperature and humidity records dynamically.
3. Perform insertion, searching, display, and deletion operations.
4. Apply linked lists to a real-world heatwave monitoring system.

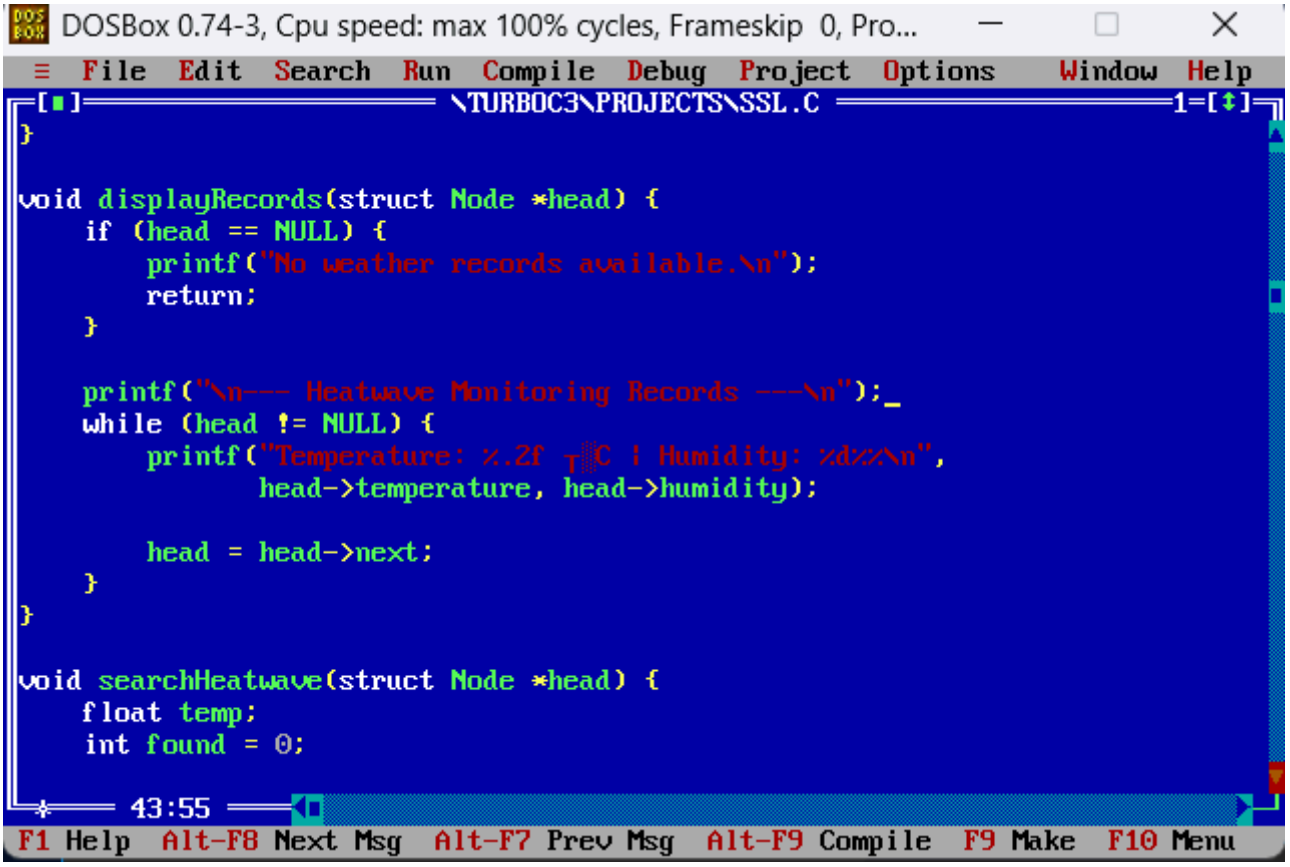
Theory:A Singly Linked List is a dynamic data structure made up of nodes. Each node contains data and a pointer to the next node. In this application, each node stores temperature and humidity readings. The list can be modified dynamically by adding or removing records.

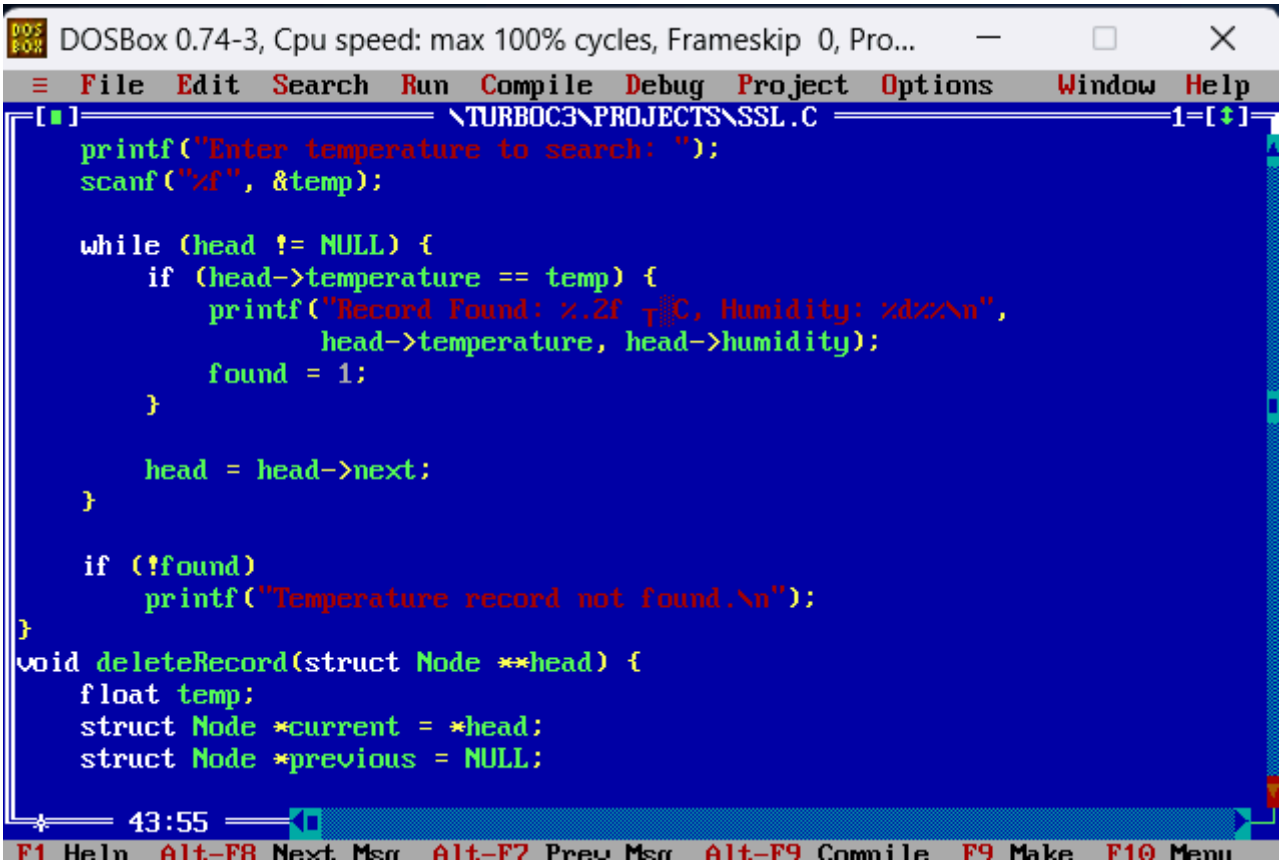
Code:

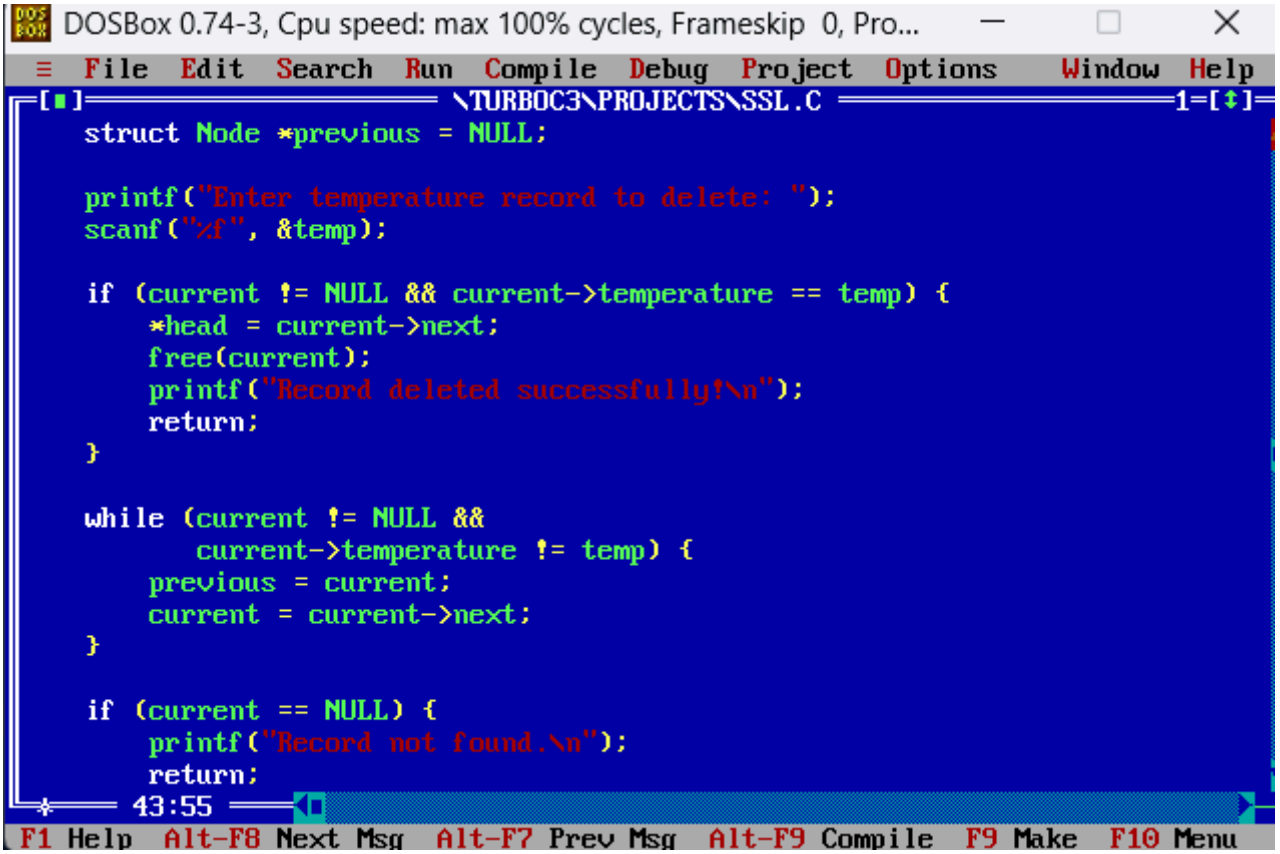

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
#include <stdio.h>
#include <stdlib.h>
struct Node {
float temperature;
int humidity;
struct Node *next;
};
void insertRecord(struct Node **head) {
struct Node *newNode, *temp;
newNode = (struct Node *)malloc(sizeof(struct Node));
printf("Enter Temperature (°C): ");
scanf("%f", &newNode->temperature);
printf("Enter Humidity (%): ");
scanf("%d", &newNode->humidity);
newNode->next = NULL;
1:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

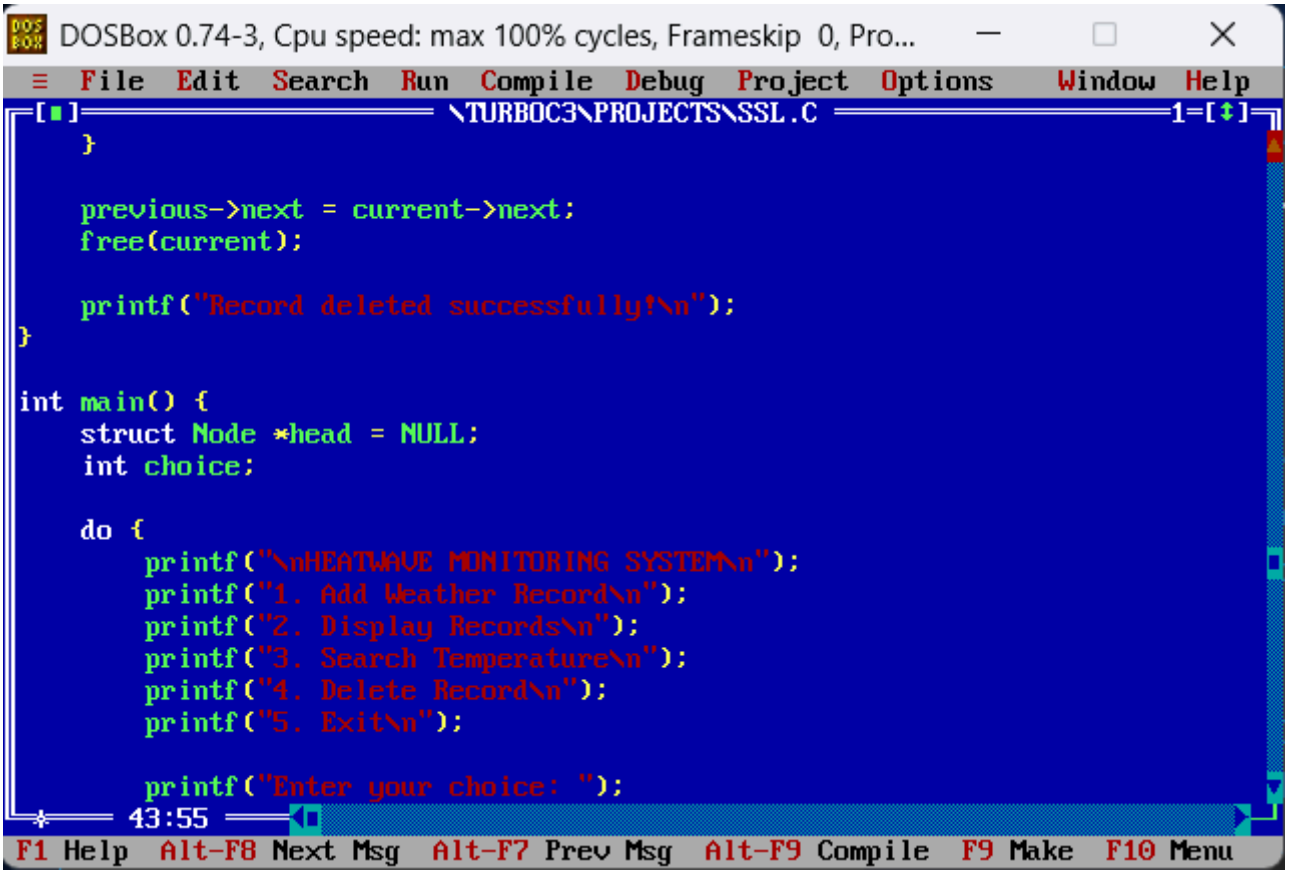

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
if (*head == NULL) {
*head = newNode;
} else {
temp = *head;
while (temp->next != NULL)
temp = temp->next;
temp->next = newNode;
}
printf("Weather record added successfully!\n");
}
void displayRecords(struct Node *head) {
if (head == NULL) {
printf("No weather records available.\n");
return;
}
printf("\n--- Heatwave Monitoring Records ---\n");
1:1
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

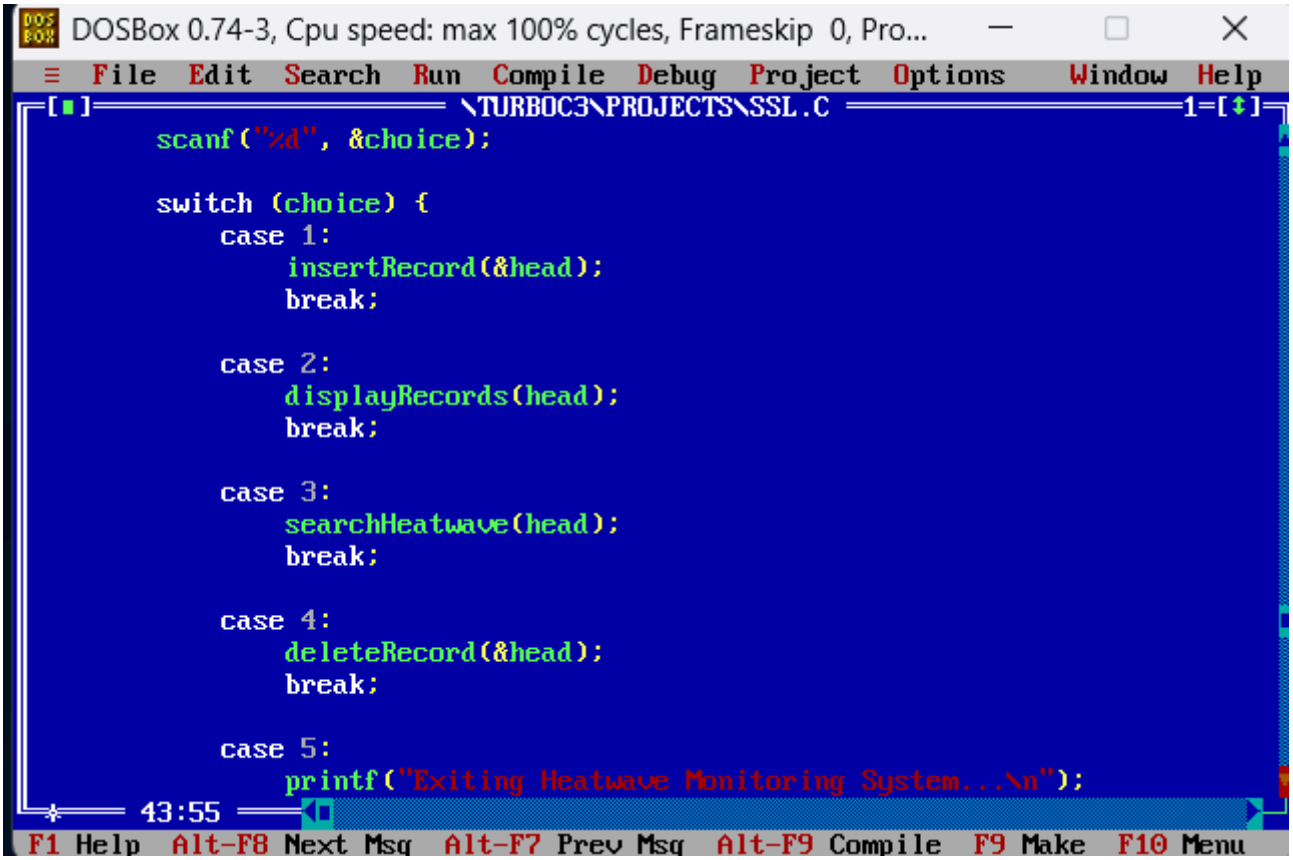

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
1=[↕]
while (head != NULL) {
printf("Temperature: %.2f °C | Humidity: %d%%\n",
head->temperature, head->humidity);
head = head->next;
}
}
void searchHeatwave(struct Node *head) {
float temp;
int found = 0;
printf("Enter temperature to search: ");
scanf("%f", &temp);
while (head != NULL) {
if (head->temperature == temp) {
printf("Record Found: %.2f °C, Humidity: %d%%\n",
head->temperature, head->humidity);
found = 1;
}
43:78

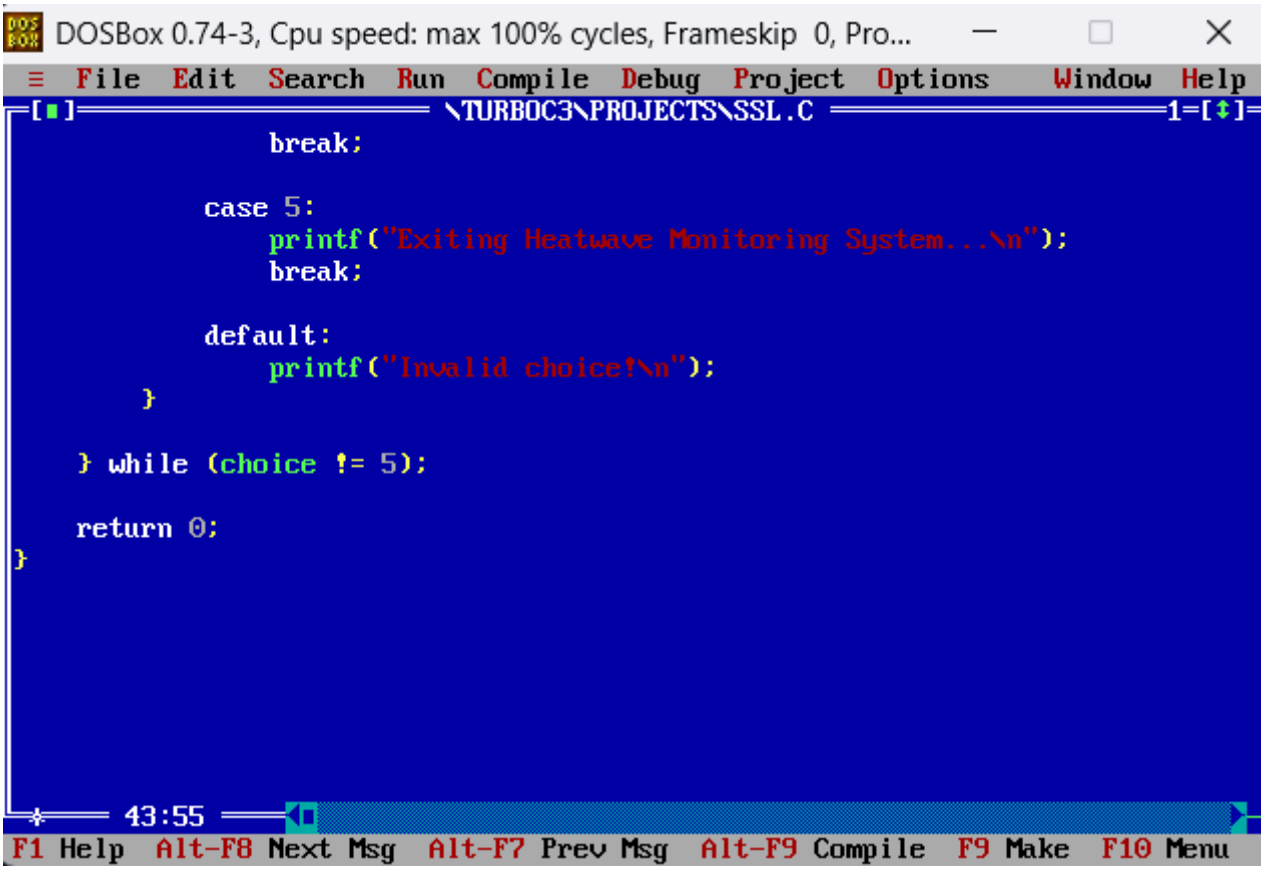

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
}
void displayRecords(struct Node *head) {
if (head == NULL) {
printf("No weather records available.\n");
return;
}
printf("\n--- Heatwave Monitoring Records ---\n");
while (head != NULL) {
printf("Temperature: %.2f °C | Humidity: %d%%\n",
head->temperature, head->humidity);
head = head->next;
}
}
void searchHeatwave(struct Node *head) {
float temp;
int found = 0;
43:55
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
printf("Enter temperature to search: ");
scanf("%f", &temp);
while (head != NULL) {
if (head->temperature == temp) {
printf("Record Found: %.2f °C, Humidity: %d%%\n",
head->temperature, head->humidity);
found = 1;
}
head = head->next;
}
if (!found)
printf("Temperature record not found.\n");
}
void deleteRecord(struct Node **head) {
float temp;
struct Node *current = *head;
struct Node *previous = NULL;
43:55
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

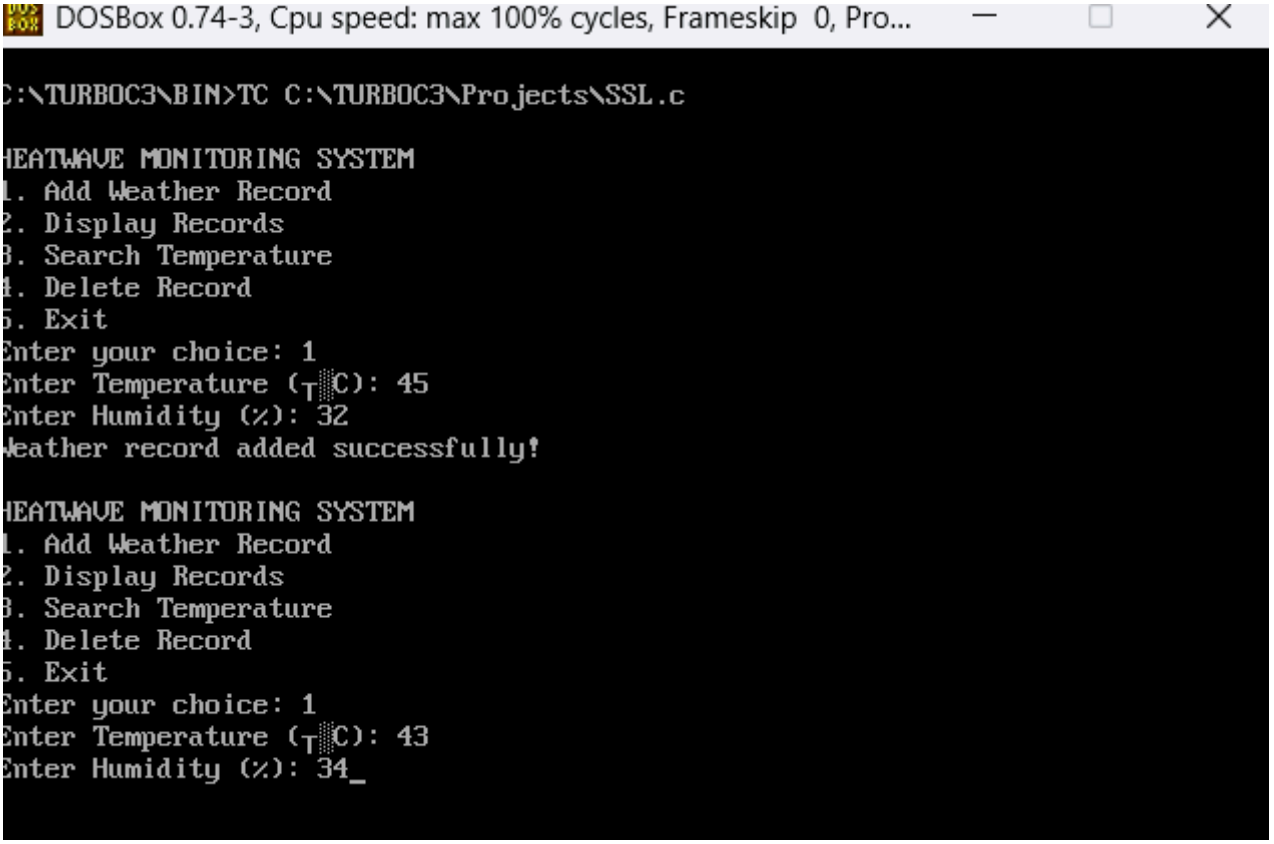

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
struct Node *previous = NULL;
printf("Enter temperature record to delete: ");
scanf("%f", &temp);
if (current != NULL && current->temperature == temp) {
*head = current->next;
free(current);
printf("Record deleted successfully!\n");
return;
}
while (current != NULL &&
current->temperature != temp) {
previous = current;
current = current->next;
}
if (current == NULL) {
printf("Record not found.\n");
return;
43:55
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

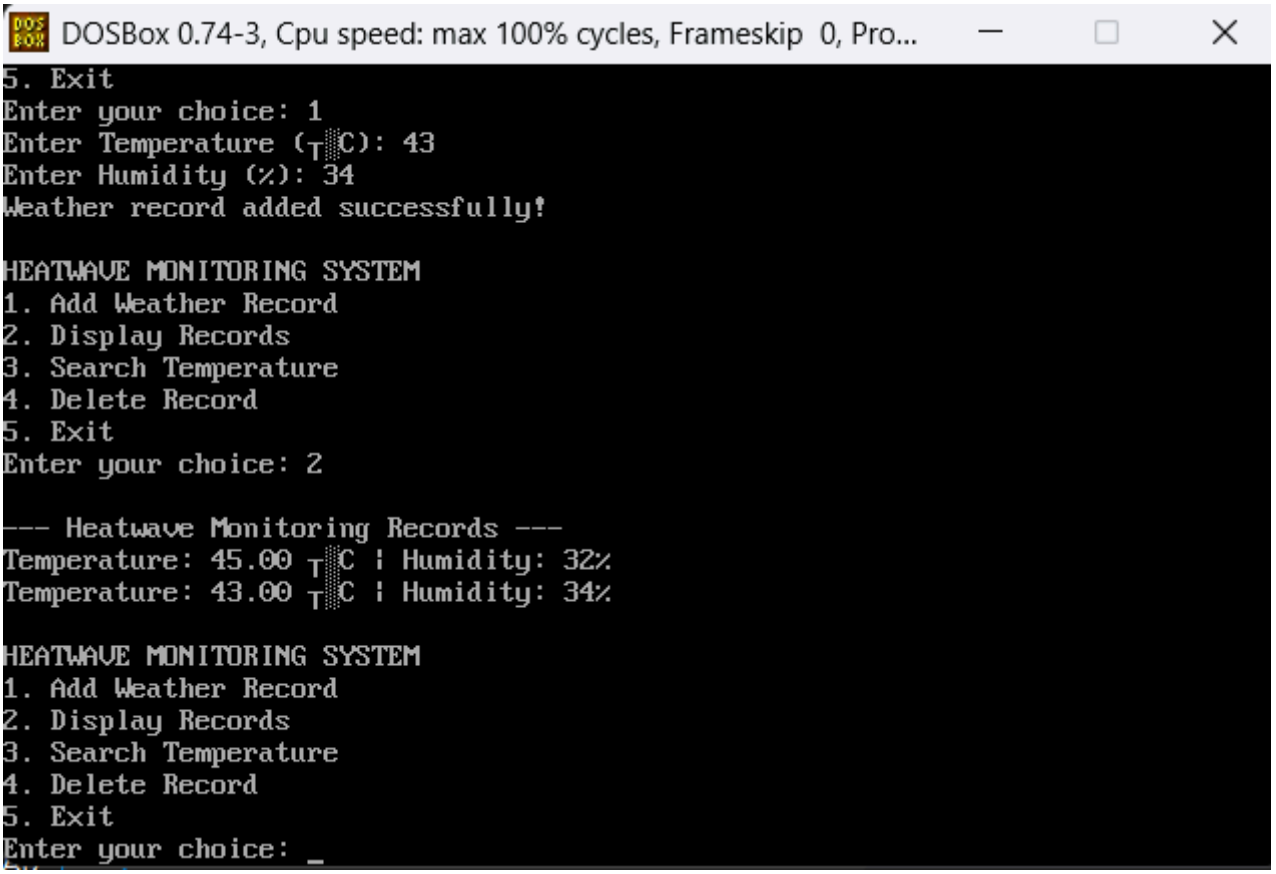

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
}
previous->next = current->next;
free(current);
printf("Record deleted successfully!\n");
}
int main() {
struct Node *head = NULL;
int choice;
do {
printf("\nHEATWAVE MONITORING SYSTEM\n");
printf("1. Add Weather Record\n");
printf("2. Display Records\n");
printf("3. Search Temperature\n");
printf("4. Delete Record\n");
printf("5. Exit\n");
printf("Enter your choice: ");
43:55
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

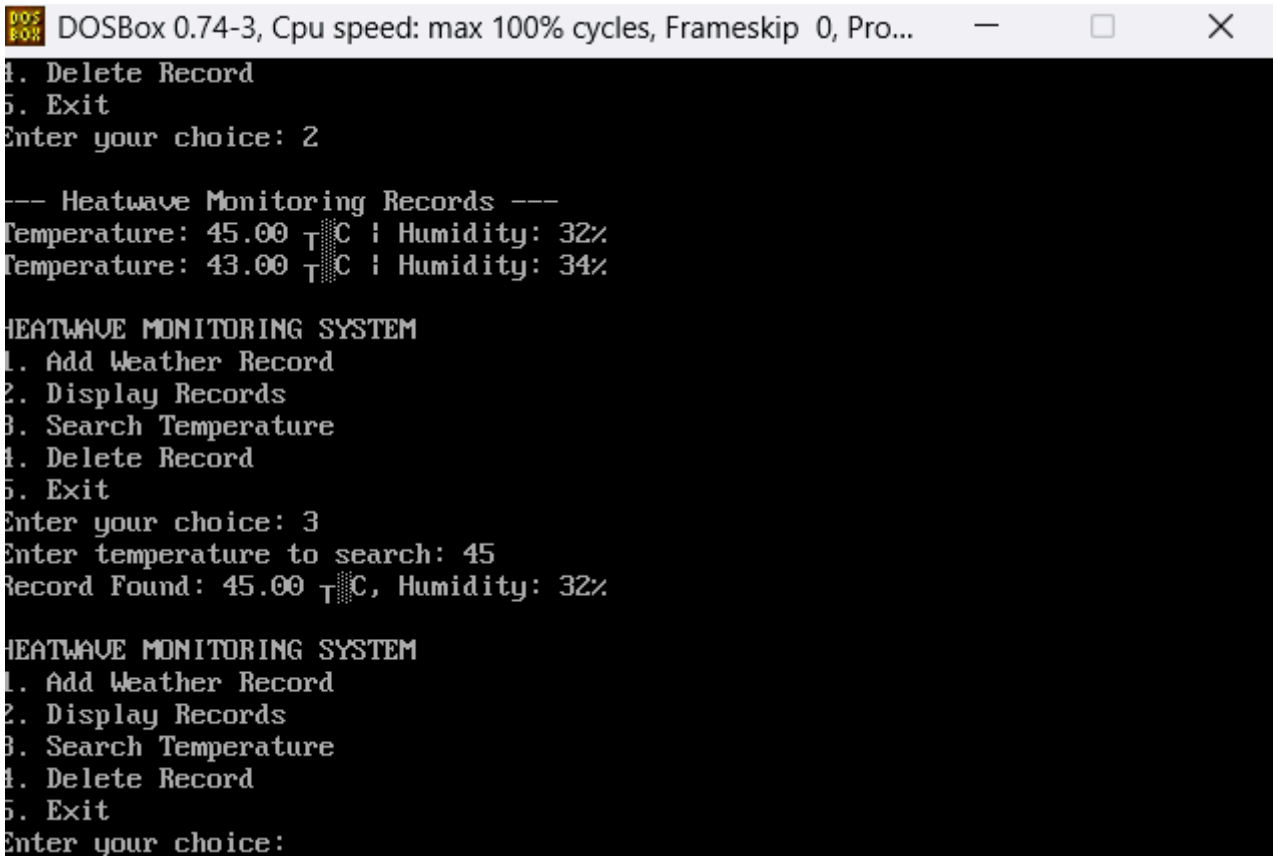

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
scanf("%d", &choice);
switch (choice) {
case 1:
insertRecord(&head);
break;
case 2:
displayRecords(head);
break;
case 3:
searchHeatwave(head);
break;
case 4:
deleteRecord(&head);
break;
case 5:
printf("Exiting Heatwave Monitoring System...\n");
43:55
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

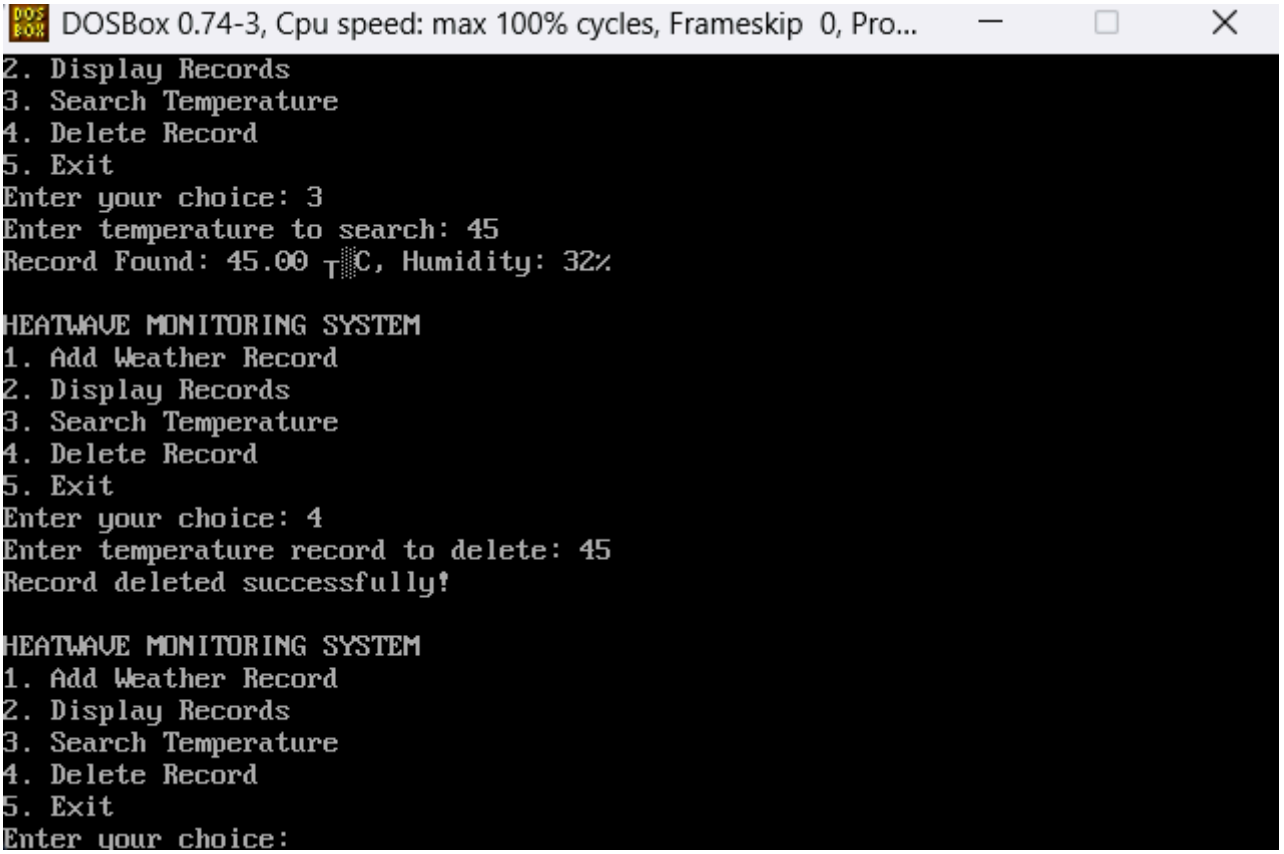

DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
File Edit Search Run Compile Debug Project Options Window Help
\TURBOC3\PROJECTS\SSL.C
break;
case 5:
printf("Exiting Heatwave Monitoring System...\n");
break;
default:
printf("Invalid choice!\n");
}
} while (choice != 5);
return 0;
}
43:55
F1 Help Alt-F8 Next Msg Alt-F7 Prev Msg Alt-F9 Compile F9 Make F10 Menu

Output:


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
C:\TURBOC3\BIN>TC C:\TURBOC3\Projects\SSL.c
HEATWAVE MONITORING SYSTEM
1. Add Weather Record
2. Display Records
3. Search Temperature
4. Delete Record
5. Exit
Enter your choice: 1
Enter Temperature (°C): 45
Enter Humidity (%): 32
Weather record added successfully!
HEATWAVE MONITORING SYSTEM
1. Add Weather Record
2. Display Records
3. Search Temperature
4. Delete Record
5. Exit
Enter your choice: 1
Enter Temperature (°C): 43
Enter Humidity (%): 34_


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
5. Exit
Enter your choice: 1
Enter Temperature (°C): 43
Enter Humidity (%): 34
Weather record added successfully!
HEATWAVE MONITORING SYSTEM
1. Add Weather Record
2. Display Records
3. Search Temperature
4. Delete Record
5. Exit
Enter your choice: 2
--- Heatwave Monitoring Records ---
Temperature: 45.00 °C | Humidity: 32%
Temperature: 43.00 °C | Humidity: 34%
HEATWAVE MONITORING SYSTEM
1. Add Weather Record
2. Display Records
3. Search Temperature
4. Delete Record
5. Exit
Enter your choice:


DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 0, Pro...
4. Delete Record
5. Exit
Enter your choice: 2
--- Heatwave Monitoring Records ---
Temperature: 45.00 °C | Humidity: 32%
Temperature: 43.00 °C | Humidity: 34%
HEATWAVE MONITORING SYSTEM
1. Add Weather Record
2. Display Records
3. Search Temperature
4. Delete Record
5. Exit
Enter your choice: 3
Enter temperature to search: 45
Record Found: 45.00 °C, Humidity: 32%
HEATWAVE MONITORING SYSTEM
1. Add Weather Record
2. Display Records
3. Search Temperature
4. Delete Record
5. Exit
Enter your choice:



Outcome: Successfully implemented a singly linked list to store and manage heatwave temperature and humidity records using basic linked-list operations.

Conclusion: The experiment demonstrates the practical use of a singly linked list for managing dynamically changing heatwave monitoring data.